

MAT 127A Discussion 2 Notes

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§I Image and Preimage

Recall that the **image** of a subset $A \subseteq X$ under a function $f : X \rightarrow Y$ is defined as

$$f(A) = \{y \in Y \mid y = f(x) \text{ for some } x \in A\}.$$

When we see set builder notation like this, we should automatically generate this proposition:

$$y \in f(A) \text{ if and only if } y = f(x) \text{ for some } x \in A.$$

Remark: This is the proposition that we should use when writing proofs, and the intermediate step that

$$y \in f(A) \text{ means } y \in \{z \in Y \mid z = f(x) \text{ for some } x \in A\},$$

which means $y = f(x)$ for some $x \in A$

should be skipped to reduce our mental load.

The **preimage** of a subset $C \subseteq Y$ under a function $f : X \rightarrow Y$ is defined as

$$f^{-1}(C) = \{x \in X \mid f(x) \in C\}.$$

Again, we should automatically generate this proposition:

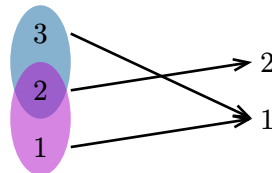
$$x \in f^{-1}(C) \text{ if and only if } f(x) \in C.$$

Decide (no need to prove) whether each of the following statements is always true for a general function $f : X \rightarrow Y$ and subsets $A, B \subseteq X$ and $C, D \subseteq Y$. If not, demonstrate a counterexample.

- | | |
|---|-------------------------------------|
| (1) $f(A \cup B) = f(A) \cup f(B)$ | (2) $f(\emptyset) = \emptyset$ |
| (3) $f(A \cap B) = f(A) \cap f(B)$ | (4) $f(X) = Y$ |
| (5) $f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D)$ | (6) $f^{-1}(\emptyset) = \emptyset$ |
| (7) $f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$ | (8) $f^{-1}(Y) = X$ |

Solution. (1, 2, 5, 6, 7, 8) are always true.

- (3) is not always true. As a concrete example, consider $X = \{1, 2, 3\}$ and $Y = \{1, 2\}$ with $f(1) = 1$, $f(2) = 2$, and $f(3) = 1$.



Then,

$$f(\{1, 2\} \cap \{2, 3\}) = f(\{1\}) = \{1\},$$

but

$$f(\{1, 2\}) \cap f(\{2, 3\}) = \{1, 2\} \cap \{1, 2\} = \{1, 2\}.$$

The key to this example are the elements 1 and 3 which do not lie in $\{1, 2\} \cap \{2, 3\}$, but f maps both of them to 1 — information that is lost if we take the intersection first.

- (4) is true if and only if f is surjective. Any non-surjective function is a counterexample.

Remark: The facts that image does not necessarily preserve binary intersection and the full subset (a.k.a. improper subset) are segments of the catchphrase that “image does not play well with intersection.” The relationship between the full subset and the intersection operation is that

the full subset is the identity for intersection,

meaning that $A \cap X = X \cap A = A$. This is completely analogous to how 0 is the identity for addition because $a + 0 = 0 + a = a$, and 1 is the identity for multiplication because $a \cdot 1 = 1 \cdot a = a$. Similarly, we can say that the empty set is the identity for union. This is the reason why I placed empty set next to union in the same rows, and full set next to intersection in the same rows.

§II Cardinality

Recall that a set X is **finite** if

$$X \approx \{1, 2, \dots, n\} \text{ for some } n \in \mathbb{N} \text{ or } X = \emptyset.$$

The $\approx$ relation here means that there exists a bijection between the two sets.

A set X is **infinite** if it is not finite. A useful alternative characterization is that there exists a sequence of distinct elements $x_1, x_2, \dots \in X$.

A set X is **countable** if $X \approx \mathbb{N}$ or that X is finite. A useful alternative characterization is that X is countable if and only if X is empty or that there exists a surjection from $\mathbb{N}$ to X .

Remark: Yet, another useful characterization is that X is countable if and only if there exists an injection from X to $\mathbb{N}$, but we wouldn't be using this in this discussion.

Since a function from $\mathbb{N}$ to X is just a sequence, we can leverage this to build intuition about countable sets. If X is nonempty, then it is countable if and only if there exists a sequence $x_1, x_2, \dots \in X$ such that $X = \{x_1, x_2, \dots\}$. We can see the function $x_\bullet$ as a function $\mathbb{N} \rightarrow X$.

(Lecture 10.03.25) Let X be a set, and let $A_1, A_2, \dots$ be a sequence of nonempty countable subsets of X .

- (a) Show that $A_1 \cup A_2$ is countable.

Hint: Write $A_1 = \{x_1, x_2, \dots\}$ and $A_2 = \{y_1, y_2, \dots\}$, can we write down a sequence $z_1, z_2, \dots \in A_1 \cup A_2$ such that $A_1 \cup A_2 = \{z_1, z_2, \dots\}$?

- (b) Show that $\bigcup_{i=1}^{\infty} A_i$ is countable.

You may directly use the fact that $\mathbb{N} \approx \mathbb{N} \times \mathbb{N}$ if you can describe a construction of the bijection but you do not need to prove that it is one.

Proof.

- (a) Since A_1 and A_2 are nonempty and countable, we can write them as

$$A_1 = \{x_1, x_2, \dots\} \text{ and } A_2 = \{y_1, y_2, \dots\}.$$

It suffices to show that there exists a sequence $z_1, z_2, \dots \in A_1 \cup A_2$ such that $A_1 \cup A_2 = \{z_1, z_2, \dots\}$. The idea is to write

$$A_1 \cup A_2 = \{x_1, y_1, x_2, y_2, x_3, y_3, \dots\}.$$

Through pattern matching, we see that we can define such a sequence by

$$z_{2n-1} = x_n \text{ and } z_{2n} = y_n \text{ for all } n \in \mathbb{N}.$$

It remains to show that $A_1 \cup A_2 = \{z_1, z_2, \dots\}$. Fix an arbitrary $a \in A_1 \cup A_2$, we wish to show that $a = z_n$ for some $n \in \mathbb{N}$. By the definition of union, we have $a \in A_1$ or $a \in A_2$.

- If $a \in A_1$, then $a = x_n$ for some $n \in \mathbb{N}$. Thus, $a = z_{2n-1}$.
- If $a \in A_2$, then $a = y_n$ for some $n \in \mathbb{N}$. Thus, $a = z_{2n}$.

This shows that $A_1 \cup A_2 = \{z_1, z_2, \dots\}$.

Remark: Really, this just shows that $A_1 \cup A_2 \subseteq \{z_1, z_2, \dots\}$, but the other direction is trivial since $z_1, z_2, \dots$ is a sequence in $A_1 \cup A_2$.

As an alternative solution, we can also deploy the language of functions as opposed to sequences. Since A_1 and A_2 are nonempty and countable, there exist surjections $f : \mathbb{N} \rightarrow A_1$ and $g : \mathbb{N} \rightarrow A_2$. It suffices to show that there exists a surjection $h : \mathbb{N} \rightarrow A_1 \cup A_2$. Like what we did for sequences, the idea is to define

$$h(2n-1) = f(n) \text{ and } h(2n) = g(n) \text{ for all } n \in \mathbb{N}.$$

To show that this is a surjection, fix an arbitrary $a \in A_1 \cup A_2$, we wish to show that $a = h(n)$ for some $n \in \mathbb{N}$. By the definition of union, we have $a \in A_1$ or $a \in A_2$.

- If $a \in A_1$, then $a = f(n)$ for some $n \in \mathbb{N}$. Thus, $a = h(2n-1)$.
- If $a \in A_2$, then $a = g(n)$ for some $n \in \mathbb{N}$. Thus, $a = h(2n)$.

This shows that h is a surjection.

Remark: You see, this is the exact same proof written differently under the correspondences $f \leftrightarrow x_\bullet$, $g \leftrightarrow y_\bullet$, and $h \leftrightarrow z_\bullet$.

- (b) If we use $x_\bullet$ for A_1 , $y_\bullet$ for A_2 , and $z_\bullet$ for A_3 , and so on, we will run out of letters (some may say we already ran out of letters). Instead, we have to use indices. Since A_i is nonempty and countable for each $i \in \mathbb{N}$, we can write

$$A_i = \{x_{(i,1)}, x_{(i,2)}, \dots\} \text{ for all } i \in \mathbb{N}.$$

The question is how to find a sequence $z_1, z_2, \dots \in \bigcup_{i=1}^{\infty} A_i$ that goes through the grid of elements $x_{(i,j)}$ over $(i,j) \in \mathbb{N} \times \mathbb{N}$. This is where the fact that $\mathbb{N} \approx \mathbb{N} \times \mathbb{N}$ comes in handy. It tells us that there is a bijection $h : \mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}$, and we can simply define

$$z_n = x_{h(n)}.$$

We indeed have $\bigcup_{i=1}^{\infty} A_i = \{z_1, z_2, \dots\}$ because for any arbitrary $a \in \bigcup_{i=1}^{\infty} A_i$, we have $a \in A_i$ for some $i \in \mathbb{N}$, which tells us that $a = x_{(i,j)}$ for some $j \in \mathbb{N}$. Since h is a surjection, there exists some $n \in \mathbb{N}$ such that $h(n) = (i,j)$, and therefore $a = x_{(i,j)} = x_{h(n)} = z_n$.

A construction of h can be described by first breaking down $\mathbb{N} \times \mathbb{N}$ into subsets whose elements sum to the same value:

$$\begin{aligned}
D_2 &= \{(1, 1)\}, \\
D_3 &= \{(1, 2), (2, 1)\}, \\
D_4 &= \{(1, 3), (2, 2), (3, 1)\}, \\
D_5 &= \{(1, 4), (2, 3), (3, 2), (4, 1)\}, \\
&\vdots \\
D_n &= \{(1, n-1), (2, n-2), \dots, (n-1, 1)\}, \\
&\vdots
\end{aligned}$$

Then, simply enumerate the elements in $D_2, D_3, D_4, \dots$ in that order will yield a bijection from $\mathbb{N}$ to $\mathbb{N} \times \mathbb{N}$.

Remark: I called them D_n because they look like the diagonals in the $\mathbb{N} \times \mathbb{N}$ grid. In the lecture, they are literally referred to as diagonals.

Additional exercise: It is your turn to write a proof based on functions instead of sequences!

(Case study: why countability matters in real analysis?) Let A be a set consists of positive real numbers. We say A is **summable** if there exists an upper bound on the sums of finite subsets of A , i.e., there exists some $M \in \mathbb{R}$ such that for every finite subset $\{a_1, \dots, a_n\} \subseteq A$, we have $a_1 + \dots + a_n \leq M$. (The least upper bound is the sum of A , but this is not important here.)

- (a) Find an infinite summable set of positive real numbers.
- (b) Show that if a set A of positive real numbers is summable, then A is countable.

Proof.

- (a) $A = \{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \dots\}$ is summable. To see this, consider an arbitrary finite subset $\{a_1, a_2, \dots, a_n\} \subseteq A$, and without loss of generality, assume we write them in descending order, i.e., $a_1 > a_2 > \dots > a_n$. Then, we must have

$$a_1 \leq \frac{1}{2}, \quad a_2 \leq \frac{1}{4}, \quad \dots, \quad a_n \leq \frac{1}{2^n}.$$

Therefore, by the geometric sum, we have

$$a_1 + a_2 + \dots + a_n \leq \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2^n} = 1 - \frac{1}{2^n} < 1.$$

Thus, A is summable with $M = 1$. (In fact, the sum of A is 1.)

- (b) Note that we can break A into countably many subsets:

$$\begin{aligned}
A_1 &= A \cap \left[\frac{1}{2}, \infty\right), \\
A_2 &= A \cap \left[\frac{1}{4}, \frac{1}{2}\right), \\
A_3 &= A \cap \left[\frac{1}{8}, \frac{1}{4}\right), \\
A_4 &= A \cap \left[\frac{1}{16}, \frac{1}{8}\right), \\
&\vdots \\
A_n &= A \cap \left[\frac{1}{2^n}, \frac{1}{2^{n-1}}\right), \\
&\vdots
\end{aligned}$$

Since A is summable, let us say it is summable with an upper bound M , then A_n must be summable by the same upper bound. This implies that each A_n is finite because otherwise, we could find a sequence of distinct elements $a_1, a_2, \dots \in A_n$, but then $a_i \geq \frac{1}{2^n}$ implies that

$$a_1 + a_2 + \dots + a_k \geq \frac{k}{2^n}$$

which can be made arbitrarily large, contradicting the summability of A_n (we have $a_1 + \dots + a_k > M$ when $k > 2^n M$). Thus, $A = \bigcup_{n=1}^{\infty} A_n$ is a union of countably many finite sets, and therefore countable by the previous exercise (or by enumerating the elements in the order $A_1, A_2, A_3, \dots$).

Remark: Defining $A_n = A \cap [\frac{1}{2^n}, \infty)$ uniformly (as opposed to making A_1 an exception) also works. We do not need the A_n 's to be disjoint.